

FIRST ORDER DIFFERENTIAL EQUATIONS SOLUTION

Part A — Separable Differential Equations

1. $\frac{dy}{dx} = xy$

Step 1: Separate variables:

$$\frac{1}{y} dy = x dx.$$

Step 2: Integrate both sides:

$$\int \frac{1}{y} dy = \int x dx \quad \Rightarrow \quad \ln |y| = \frac{x^2}{2} + C.$$

Step 3: Solve for y :

$$y = C_1 e^{x^2/2}, \quad C_1 = \pm e^C.$$

2. $\frac{dy}{dx} = \frac{y}{x}, y(1) = 3$

Step 1: Separate:

$$\frac{1}{y} dy = \frac{1}{x} dx.$$

Step 2: Integrate:

$$\ln |y| = \ln |x| + C \Rightarrow y = C_1 x.$$

Step 3: Use initial condition $y(1) = 3$:

$$3 = C_1 \cdot 1 \Rightarrow C_1 = 3.$$

Solution: $y = 3x$.

$$3. \frac{dy}{dx} = \frac{x^2}{1+y^2}$$

Step 1: Separate:

$$(1+y^2) dy = x^2 dx.$$

Step 2: Integrate:

$$\int (1+y^2) dy = \int x^2 dx \Rightarrow y + \frac{y^3}{3} = \frac{x^3}{3} + C.$$

Solution: Implicit solution $y + \frac{y^3}{3} = \frac{x^3}{3} + C$. (Can't solve for y in elementary closed form.)

$$4. (1+y^2) dy = x dx$$

(This is same form as Q3 but arranged differently.)

Step 1: Integrate both sides:

$$\int (1+y^2) dy = \int x dx \Rightarrow y + \frac{y^3}{3} = \frac{x^2}{2} + C.$$

Solution: $y + \frac{y^3}{3} = \frac{x^2}{2} + C$.

$$5. \text{Growth model } \frac{dP}{dt} = 5P$$

Step 1: Separate:

$$\frac{1}{P} dP = 5 dt.$$

Step 2: Integrate:

$$\ln |P| = 5t + C \Rightarrow P = C_1 e^{5t}.$$

Solution: $P(t) = P(0) e^{5t}$ if initial value known.

Part B — Linear First-Order DEs

A linear first-order DE has form $y' + p(x)y = q(x)$. Use integrating factor $\mu(x) = e^{\int p(x) dx}$.

6. $\frac{dy}{dx} + 3y = 6$

Step 1: $p(x) = 3$. Integrating factor $\mu = e^{3x}$.

Step 2: Multiply: $(e^{3x}y)' = 6e^{3x}$.

Step 3: Integrate:

$$e^{3x}y = \int 6e^{3x} dx = 2e^{3x} + C.$$

Step 4: Solve:

$$y = 2 + Ce^{-3x}.$$

7. $\frac{dy}{dx} - 2y = e^{3x}, y(0) = 1$

Step 1: $p(x) = -2$. $\mu = e^{\int -2 dx} = e^{-2x}$.

Step 2: Multiply: $(e^{-2x}y)' = e^{-2x}e^{3x} = e^x$.

Step 3: Integrate:

$$e^{-2x}y = \int e^x dx = e^x + C.$$

Step 4: Solve:

$$y = e^{2x}(e^x + C) = e^{3x} + Ce^{2x}.$$

Step 5: Apply $y(0) = 1$:

$$1 = e^0 + Ce^0 \Rightarrow 1 = 1 + C \Rightarrow C = 0.$$

Solution: $y = e^{3x}$.

$$8. x \frac{dy}{dx} + y = x^2$$

First rewrite in standard linear form: $\frac{dy}{dx} + \frac{1}{x}y = x$, valid for $x \neq 0$.

Step 1: $p(x) = 1/x$. $\mu = e^{\int 1/x \, dx} = e^{\ln x} = x$.

Step 2: Multiply: $(xy)' = x \cdot x = x^2$.

Step 3: Integrate:

$$xy = \int x^2 \, dx = \frac{x^3}{3} + C.$$

Step 4: Solve:

$$y = \frac{x^2}{3} + \frac{C}{x}.$$

$$9. \frac{dy}{dx} + \frac{1}{x}y = 4x$$

Step 1: $p(x) = 1/x$. $\mu = x$.

Step 2: $(xy)' = x \cdot 4x = 4x^2$.

Step 3: Integrate:

$$xy = \int 4x^2 \, dx = \frac{4x^3}{3} + C.$$

Step 4: Solve:

$$y = \frac{4x^2}{3} + \frac{C}{x}.$$

$$10. \frac{dy}{dx} + 2xy = x$$

This is linear with $p(x) = 2x$. $\mu = e^{\int 2x \, dx} = e^{x^2}$.

Step 1: Multiply: $(e^{x^2}y)' = xe^{x^2}$.

Step 2: Integrate RHS: $\int xe^{x^2} \, dx$. Let $u = x^2$, $du = 2x \, dx$ so $\int xe^{x^2} \, dx = \frac{1}{2}e^{x^2} + C$.

Step 3: So

$$e^{x^2}y = \frac{1}{2}e^{x^2} + C \Rightarrow y = \frac{1}{2} + Ce^{-x^2}.$$

Part C — Exact Equations

A differential $M(x, y) \, dx + N(x, y) \, dy = 0$ is exact if $M_y = N_x$ (in a simply connected region).

11. $(2xy + 3) dx + (x^2 + 2y) dy = 0$

Step 1: Identify $M = 2xy + 3$, $N = x^2 + 2y$.

Step 2: Compute partials:

$$M_y = 2x, \quad N_x = 2x \Rightarrow M_y = N_x.$$

Exact.

Step 3: Find potential $\Phi(x, y)$ with $\Phi_x = M$, $\Phi_y = N$.

Integrate M w.r.t x :

$$\Phi(x, y) = \int (2xy + 3) dx = x^2y + 3x + h(y).$$

Differentiate w.r.t y :

$$\Phi_y = x^2 + h'(y) \stackrel{!}{=} N = x^2 + 2y \Rightarrow h'(y) = 2y \Rightarrow h(y) = y^2 + C.$$

Solution (implicit):

$$\Phi(x, y) = x^2y + 3x + y^2 = C.$$

13. $(2xy + y^2) dx + (x^2 + xy) dy = 0$

Step 1: $M = 2xy + y^2$, $N = x^2 + xy$.

Step 2: Compute $M_y = 2x + 2y$, $N_x = 2x + y$. They are **not equal**, so not exact.

Step 3: Look for integrating factor depending only on x (test):

$$\mu(x) \text{ exists if } \frac{M_y - N_x}{N} \text{ is function of } x \text{ only.}$$

Compute:

$$\frac{M_y - N_x}{N} = \frac{(2x + 2y) - (2x + y)}{x^2 + xy} = \frac{y}{x(x + y)} = \frac{1}{x} \cdot \frac{y}{x + y},$$

which depends on y as well — not function of x only.

Test integrating factor depending on y :

$$\frac{N_x - M_y}{M} = \frac{(2x + y) - (2x + 2y)}{2xy + y^2} = \frac{-y}{y(2x + y)} = \frac{-1}{2x + y}.$$

This depends on both x and y through $2x + y$, but it is a function of the combination $2x + y$ — not purely y . So no simple $\mu(x)$ or $\mu(y)$ from these tests.



Step 4: But notice M and N are homogeneous of degree 2. For homogeneous equations, try integrating factor $\mu = 1/(x^2)$ or $1/y^2$ etc. Try $\mu = 1/y^2$:

$$\tilde{M} = \frac{2xy + y^2}{y^2} = 2\frac{x}{y} + 1, \quad \tilde{N} = \frac{x^2 + xy}{y^2} = \left(\frac{x}{y}\right)^2 + \frac{x}{y}.$$

Set $u = x/y$ — this transforms to a separable form; or more systematically one can use $\mu = 1/(xy)$ sometimes. (Multiple methods exist; for class you can show the substitution $u = x/y$ to reduce to separable).

One method (substitution): Let $u = x/y \Rightarrow x = uy$. Then compute dy/dx etc — this transforms equation into separable ODE. (Because this question asked to "find integrating factor depending on x "—we found none; show alternative substitution.)

Part D — Integrating Factor (Non-Exact Linear Equations)

14. $\frac{dy}{dx} + \frac{2}{x}y = 3x^2$

Step 1: Linear with $p(x) = 2/x$. $\mu = e^{\int 2/x dx} = e^{2 \ln x} = x^2$ (for $x > 0$).

Step 2: Multiply:

$$(x^2 y)' = x^2 \cdot 3x^2 = 3x^4.$$

Step 3: Integrate:

$$x^2 y = \int 3x^4 dx = \frac{3x^5}{5} + C.$$

Step 4: Solve:

$$y = \frac{3x^3}{5} + \frac{C}{x^2}.$$

15. $\frac{dy}{dx} - y \cot x = \sin x$

Step 1: $p(x) = -\cot x$. $\mu = e^{\int -\cot x \, dx} = e^{-\ln(\sin x)} = \frac{1}{\sin x}$. (Assume $\sin x \neq 0$.)

Step 2: Multiply:

$$\left(\frac{y}{\sin x}\right)' = \frac{\sin x}{\sin x} = 1.$$

Step 3: Integrate:

$$\frac{y}{\sin x} = x + C \Rightarrow y = (x + C) \sin x.$$

16. $\frac{dy}{dx} + 4y = e^{-2x}$, $y(0) = 0$

Step 1: $p = 4$. $\mu = e^{4x}$.

Step 2: $(e^{4x}y)' = e^{4x}e^{-2x} = e^{2x}$.

Step 3: Integrate:

$$e^{4x}y = \int e^{2x} \, dx = \frac{1}{2}e^{2x} + C.$$

Step 4: Solve:

$$y = \frac{1}{2}e^{-2x} + Ce^{-4x}.$$

Step 5: Apply initial condition $y(0) = 0$:

$$0 = \frac{1}{2} + C \Rightarrow C = -\frac{1}{2}.$$

Solution: $y = \frac{1}{2}e^{-2x} - \frac{1}{2}e^{-4x} = \frac{1}{2}(e^{-2x} - e^{-4x})$.